

Core Web Vitals: The Image Impact

What are Core Web Vitals?

Google's official metrics for measuring user experience:

1. **LCP** (Largest Contentful Paint) - Loading performance
2. **FID** (First Input Delay) - Interactivity
3. **CLS** (Cumulative Layout Shift) - Visual stability

Images are the #1 cause of poor LCP scores.

The Image Problem

Before Optimization

None

Hero Image: 4.2 MB

Thumbnail Grid: 15 × 800 KB = 12 MB

Total: 16.2 MB

LCP: 4.8 seconds ❌

Page Load: 8.2 seconds ❌

After Optimization

None

Hero Image: 95 KB (WebP)

Thumbnail Grid: 15 × 35 KB = 525 KB

Total: 620 KB

LCP: 1.2 seconds ✅

Page Load: 2.1 seconds ✅

Result: 96% size reduction, 4x faster load

The Complete Fix List

1. Compress

Goal: Reduce file size without visible quality loss

Tools:

- TinyPNG/TinyJPG (lossy, best compression)
- ImageOptim (lossless, Mac)
- Squoosh (Google, browser-based)

Targets:

- Hero images: <100 KB
- Thumbnails: <50 KB
- Icons: <10 KB

2. Modern Format: WebP

Why: 25-35% smaller than JPEG, 26% smaller than PNG

```
HTML
<picture>
  <source srcset="image.webp" type="image/webp">
  <source srcset="image.jpg" type="image/jpeg">
  
</picture>
```

Browser Support: 96%+ (Safari 14+)

Alternative: AVIF (50% smaller, but 80% support)

3. Correct Dimensions

Problem: Serving 4000px image to mobile (375px screen)

```
HTML
<!-- ❌ Bad: One size for all -->


<!-- ✅ Good: Responsive images -->

```

4. Lazy Loading

Don't load images until they're needed

HTML

```
<!-- Native lazy loading -->  
  
  
<!-- Only loads when image enters viewport -->
```

Savings: 50-70% fewer image requests on initial load

5. CDN + Automatic Optimization

Services that auto-optimize:

- Cloudflare Images
- Cloudinary
- ImageKit
- Imgix

Example (Cloudflare):

HTML

```
<!-- Original -->  
  
  
<!-- Auto-optimized by Cloudflare -->  
- Converted to WebP for supporting browsers  
- Resized to fit device  
- Cached globally
```

Measuring Impact

Google Lighthouse

```
Shell
# Run audit
npm install -g lighthouse
lighthouse https://yoursite.com --view

# Target Scores:
# Performance: 90+
# LCP: <2.5s
```

PageSpeed Insights

<https://pagespeed.web.dev/>

Before:

- Performance: 42/100
- LCP: 4.8s

After:

- Performance: 96/100
 - LCP: 1.2s
-

Image Format Decision Tree

```
None
Need transparency?
├─ YES
│   ├── Photo? → WebP
│   └─ Logo/Icon? → SVG (if vector) or PNG
│
└─ NO
    ├── Photo? → WebP or JPEG
    └─ Simple graphics? → SVG
```

Compression Settings

JPEG Quality

- **90-95%:** Print quality (unnecessary for web)
- **80-85%:** High quality (hero images)
- **70-75%:** Standard (general use) ✓
- **60-65%:** Low quality (thumbnails)
- **<60%:** Visible artifacts ✗

WebP Quality

- **85-90%:** High quality
 - **75-80%:** Standard ✓
 - **65-70%:** Thumbnails
-

Real-World Results

Case Study: E-commerce Site

Before Optimization:

- Total image size: 18.4 MB
- LCP: 5.2 seconds
- Bounce rate: 68%
- Mobile conversions: 1.8%

After Optimization:

- Total image size: 1.1 MB (94% reduction)
- LCP: 1.4 seconds
- Bounce rate: 42% (38% improvement)
- Mobile conversions: 3.2% (78% improvement)

Revenue Impact: +\$42,000/month

Quick Wins Checklist

- ☐ Compress all images (aim for <100KB each)
- ☐ Serve images in WebP format
- ☐ Use responsive images (srcset)
- ☐ Add loading="lazy" to below-fold images
- ☐ Set explicit width and height attributes

- ☐ Use a CDN
 - ☐ Remove unused images from codebase
-

Monitoring

Set Up Alerts

```
JavaScript
// Real User Monitoring (RUM)
new PerformanceObserver((list) => {
  for (const entry of list.getEntries()) {
    if (entry.entryType === 'largest-contentful-paint') {
      if (entry.renderTime > 2500) {
        // Alert: LCP > 2.5s
        logToAnalytics('slow_lcp', entry);
      }
    }
  }
}).observe({ entryTypes: ['largest-contentful-paint'] });
```

Is your site slow? Devxhub guarantees 90+ Lighthouse scores.